

Chapter 111 — Underactive Detrusor

- **Impaired emptying with age is common in both sexes** — raised outlet resistance, weak contraction, or both. The weak contraction is **detrusor underactivity (DUA)** (International Continence Society [ICS] urodynamic term); the symptomatic condition is **underactive bladder (UAB)**.
- **Outlet resistance dominates male therapy**: benign prostatic hyperplasia (BPH) raises it with age and is the target of most treatment for male lower urinary tract symptoms (LUTS).
- **DUA mostly lacks effective treatment other than catheterisation**, preferably intermittent self-catheterisation (ISC), when the postvoid residual (PVR) is raised.
- **Prevalence in referred LUTS rises with age**; cause and mechanisms are poorly understood.

Concept — *Emptying is the product of contraction against resistance; medicine can lower resistance but cannot yet strengthen the detrusor, so DUA management is largely drainage.*

TERMINOLOGY, DEFINITIONS, AND SYMPTOMS

- **Older terms abandoned**: impaired detrusor contractility, detrusor areflexia, detrusor failure.
- **ICS 2002 definition of DUA (urodynamic)**: "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."
 - **No thresholds are specified** for reduced strength, reduced duration or prolonged emptying.
- **No contraction at all → acontractile detrusor**. Distinguish from an isolated failure to void during a urodynamic study (UDS) — the "**bashful bladder**", which is common and often matches a history of inability to void in public toilets.
- **A purely urodynamic definition is impractical** (UDS is invasive, not universally available) — the reasoning behind the overactive bladder (OAB) symptom complex. But OAB is anchored on **urgency** (variably linked to detrusor overactivity [DO]); **UAB has no pathognomonic symptom** and overlaps with OAB and bladder outlet obstruction (BOO).

SYMPTOMS OF DUA BY PHASE OF THE MICTURITION CYCLE

Phase	Symptoms
Voiding	Weak stream, intermittency, hesitancy, straining
Postmicturition	Feeling of incomplete emptying
Storage	Frequency and nocturia or loss of the normal desire to void with infrequent voiding (the opposite of urgency)
Very large PVR	Incontinence, especially during sleep

BLADDER SENSATION DETERMINES THE CLINICAL PICTURE (FIG. 111.2)

	Sensation absent or very poor	Sensation preserved
Voiding pattern	Infrequent voider	Normal interval void interval if emptying normal
If emptying poor	Incontinence, worse at night (urethral relaxation in sleep)	Frequent voider : incomplete emptying, frequency, nocturia ± incontinence

- **DUA may also be asymptomatic**. This range, and its overlap with LUTS/BPH and OAB, defeats a reliable symptom-based definition.
- **Clinically significant PVR is not agreed**: one proposal is **>40% of functional capacity** (voided volume + PVR).

- **ICS-accepted symptomatic definition** (Chapple and colleagues, 2018): "Underactive bladder is characterised by a **slow urinary stream, hesitancy and straining to void**, with or without a feeling of incomplete bladder emptying sometimes with storage symptoms."
 - Footnote 1: UAB accompanies **diverse pathophysiologies; no single distinguishing symptom** exists.
 - Footnote 2: storage symptoms (nocturia, daytime frequency, reduced filling sensation, incontinence) are varied, may be highly prevalent and **are often driven by a significant PVR**.
 - Quantitative and qualitative studies support the definition in urodynamic DUA.

Figure 111.1 — Urodynamic traces

- **(A) DUA:** male, benign prostatic enlargement, voiding LUTS; **low detrusor pressure (Pdet), prolonged contraction, prolonged voiding time**, raised PVR.
- **(B) Acontractile detrusor:** young female, back injury, on ISC; no voiding contraction.

Concept — DUA is defined by a **measurement** and UAB by **symptoms**, and no symptom maps specifically onto the measurement; **residual sensation** is what makes the same weak detrusor present as frequency in one patient and infrequent voiding in another.

EPIDEMIOLOGY

- **LUTS are highly prevalent and increase with age**; the contribution of DUA is unknown.
- **Epidemiology of LUTS (EpiLUTS)** (UK, USA, Sweden; over 40s): storage LUTS commoner in females, voiding in males, postmicturition similar. Difficulty emptying is common in older community dwellers.
- **Population prevalence of DUA cannot be established:** there is **no simple non-invasive marker**.
- **Proxies fail in males:** voiding LUTS, raised PVR and low flow can all be caused by BOO.
- **In females** straining may simply accompany OAB (frequency, small volumes); but voiding LUTS with low flow and raised PVR make **DUA likely, because female BOO is rare (2.7–8% of those referred for UDS)**.
- **In referred patients, DUA is common**, especially in the elderly (Table 111.1).

TABLE 111.1 PREVALENCE OF DETRUSOR UNDERACTIVITY IN CLINICAL STUDIES — from Osman, Chapple, Abrams et al., *Eur Urol* 2014

Population	Diagnostic criteria used	Prevalence of DUA
Males aged <50 (young men with LUTS)	Pdet at maximal flow (Pdet@Qmax) <30–45 cm H ₂ O with maximal flow rate (Qmax) <12–15 mL/s; BOO index <20 with flow <12; ICS definition	9–23% across tabulated studies; 9–28% in the body text and key points (accontractile ~5–10%)
Males, mixed/all ages	Pdet@Qmax ≤30 with Qmax ≤12; relaxed sphincter electromyography (EMG) with open membranous urethra and low flow	~10%
Males aged >65–70	Pdet@Qmax <30 with Qmax <10; bladder contractility index (BCI) <100	40–48%
Females, community/clinic (mean age ~60, or >55)	ICS definition; prolonged voiding with high residual	~14–19%
Females aged >65–70	Qmax ≤12 with Pdet@Qmax ≤10; Pdet@Qmax <30 with Qmax <10	12–13%
Institutionalised elderly (mean age ~87)	Failure to empty without abdominal straining; DHIC: involuntary contraction emptying less than half the bladder	Males ~41%; females ~38–45%

- **What the table is for:** showing that DUA is common in referred LUTS and **rises steeply with age** — to roughly half of men over 70.

- **Pattern:** highest in the **institutionalised elderly**, who often have coexisting DO — **detrusor hyperactivity with impaired contractility (DHIC)**.
- **Two ranges for young men:** the tabulated studies span 9–23%; the chapter's text quotes 9–28%, presumably drawing on studies beyond the table. Quote 9–28% as the chapter figure.
- **Trap:** every study uses **different criteria** (sex-specific, flow and pressure cut-offs, nomograms), so figures cannot be pooled or extrapolated to the general population; data are also limited by population selection and post hoc interpretation.
- **DUA commonly coexists with other lower urinary tract dysfunctions (LUTDs)** in the elderly (Fig. 111.3, largest clinical urodynamic series, patients >65): nearly half of males with DUA also had DO or BOO; nearly three-quarters of females with DUA also had DO or urodynamic stress urinary incontinence (SUI).
 - The overlap supports **UDS before surgery for LUTS in the elderly**.
- **Natural history is little known.** Men with non-neurogenic DUA ($Q_{\max} < 15$ mL/s, $P_{\det}@Q_{\max} < 40$ cm H_2O) managed conservatively for 10 years showed **minimal symptomatic or urodynamic progression**. The minority later undergoing transurethral resection of the prostate (TURP), for retention or worsening LUTS, showed no preceding fall in $Q_{\max}$.

KEY POINTS: TERMINOLOGY, DEFINITIONS, AND SYMPTOMS, AND EPIDEMIOLOGY

- DUA is a common clinical problem associated with a range of storage, voiding and postmicturition LUTS.
- Its population prevalence is unknown for lack of a non-invasive marker.
- DUA affects **9–28% of males under 50 and 48% of those over 70** undergoing UDS, and is more prevalent among the institutionalised elderly.
- In females, DUA is found in **12–45%** undergoing UDS, more so among the institutionalised elderly.
- DUA commonly coexists with other LUTDs.

Concept — *Every estimate of DUA is a referral-series, criterion-dependent figure; because BOO is common in men and rare in women, the same voiding picture points to obstruction in a man but to DUA in a woman.*

ETIOPATHOGENESIS

- **DUA occurs in both sexes and diverse groups.** Recognised causes include diabetes mellitus and cauda equina compression, but **many patients have no obvious cause**, suggesting common age-related change.
- **Age-related decline in detrusor function is likely but unproven.**
 - Rodent strip data conflict; **human strips often show no decline** in response to stimulation or agonists.
 - Clinical urodynamic data suggest an age-related fall in $P_{\det}$, flow and maximum contraction strength with a rise in PVR — but mostly in **symptomatic** subjects with progressive pathology.
 - **BOO in males** adds connective tissue and reduces smooth muscle; the same changes in **females** suggest a partial contribution of ageing itself (Fig. 111.4A–C).

MECHANISMS OF DUA (FIG. 111.5)

Mechanism class	Site of dysfunction	Result	Major aetiological factors
Myogenic	Detrusor myocyte; extracellular matrix (ECM)	Loss of intrinsic contractility	Diabetes mellitus; ?BOO due to BPH; ?normative ageing
Neurogenic — efferent	Sacral cord, sacral and pelvic nerves, postganglionic neurons	Impaired activation of the detrusor	Neurological disease or injury; diabetes

Mechanism class	Site of dysfunction	Result	Major aetiological factors
Neurogenic — afferent	Peripheral afferents, anterolateral white column, posterior column	Early termination of the voiding reflex	Neurological disease or injury; diabetes; ?ageing
Neurogenic — brain circuits	Pontine micturition centre (PMC), periaqueductal grey (PAG), limbic system, hypothalamus, prefrontal cortex	Failure of integration or processing	Neurological disease or injury

- **BOO and normative ageing are probable, not definite, factors** (marked "?" in the source figure).

Myogenic Factors

- **ECM change** reduces transmission of contractile force.
- **Myocyte dysfunction** — ion storage and exchange, excitation-contraction coupling, calcium storage, energy generation — gives a **weak contraction despite normal neural drive**.
- **Elbadawi ultrastructural patterns** (electron microscopy): the **degeneration pattern** — widespread disrupted myocytes plus axonal degeneration — is associated with DUA. The classification is disputed, though others report similar findings.
- **Cause or consequence?** Whether myocyte disruption causes DUA or follows another insult is unclear.

Figure 111.4 — Morphological detrusor changes

- **Light microscopy:** (A) normal; (B) male BOO — **increased muscle mass and fibrous tissue within and between bundles**; (C) asymptomatic elderly female — **similar fibrosis**, so ECM change also occurs with ageing.
- **Electron microscopy:** (D) normal bundles separated by a collagen-rich septum, fascicles by a thin microseptum; (E) normal cells — smooth intact sarcolemma, thin interstitium; (F) **degeneration pattern** — shrivelled cells, sarcolemmal breakdown, debris and collagen in the interstitium.

Neurogenic Factors

Brain Circuits

- **The micturition reflex is a spino-bulbo-spinal loop:** sacral parasympathetic nucleus → PMC → sacral cord; the PMC is modulated by the cortex, particularly the limbic system.
- **PMC "direct neurons"** fire immediately before and during reflex contractions and are silent otherwise; many project to the lumbosacral cord.
- **Human functional neuroimaging** implicates the insula, hypothalamus, PAG and PMC.
- **A lesion anywhere here — disturbing perception and integration — could cause DUA**, but site and urodynamic finding do not always correlate.

Bladder Efferent Pathways

- **Interruption in the sacral cord, sacral roots or pelvic nerves → absent or reduced contraction.**
- **Autonomic innervation of the bladder declines with normal ageing.**

Bladder and Urethral Afferent Pathways

- **Intact sensation is critical to the efferent limb.** Bladder afferents monitor volume during filling and contraction magnitude during voiding; **urethral afferents** sense flow and contraction.
- **Afferent impairment reduces or prematurely ends the micturition reflex** → loss of voiding efficiency.
- **Sensation declines with age:** higher thresholds to reach capacity in older females; reduced insular response to filling on functional magnetic resonance imaging (fMRI) in older asymptomatic people; in ageing rats, weaker urethral flow-evoked contraction needing a larger triggering volume.
- **Tissue change and sensory decline both likely contribute in the elderly.**

Smith "stiff balloon" hypothesis (afferent output is a function of wall stress):

connective-tissue deposition + smooth-muscle disruption → loss of elasticity and receptive relaxation;
connective tissue also impairs sensory innervation → stiff bladder → precipitous rise in wall stress on filling
→ **delayed first sensation, then urgency soon after** → small void → wall stress collapses → afferent drive falls → **contraction terminates prematurely** → **raised PVR**.

- **Support:** in DUA, strong desire at higher volume, **similar contractility** (Watts factor), smaller voids and higher PVR than other LUTDs.
- **Urothelium:** biopsies in DUA suggest impaired urothelial signalling and sensory transduction; causality versus effect of BOO is unresolved, and animal models are limited by different innervation.

Specific Etiologic Factors (Box 111.1)

BOX 111.1 ETIOLOGIC FACTORS

Category	Factors
Idiopathic	Normal ageing; unknown factor in younger people
Neurogenic — vascular	Stroke (early phase)
Neurogenic — degenerative	Parkinson disease; multisystem atrophy
Neurogenic — demyelinating	Multiple sclerosis
Neurogenic — peripheral neuropathy	Guillain-Barré syndrome; neurosyphilis (tabes dorsalis); herpes zoster and herpes simplex; diabetes mellitus; acquired immunodeficiency syndrome (AIDS)
Neurogenic — spinal cord and cauda equina	Intervertebral disc prolapse; cauda equina lesions; spinal cord tumours; spinal canal stenosis; spinal cord injury; sacral fracture; pelvic fracture; bilateral pudendal nerve injury
Myogenic	BOO; diabetes; pelvic arterial ischaemia
Iatrogenic	Radical pelvic surgery (radical prostatectomy, radical hysterectomy, anterior resection, abdominoperineal resection); detrusor myomectomy; intravesical phenol; radiotherapy
Functional	Fowler syndrome; dysfunctional voiding
Pharmacotherapy	Anticholinergic-effect drugs (antimuscarinics, antihistamines, antipsychotics, anti-Parkinson drugs, antispasmodics, tricyclic antidepressants); opioids

- **Purpose:** history checklist — above all the **drug history**, the readily reversible category.
- **Trap: diabetes appears twice** (neuropathy and myogenic) because it acts by both routes; stroke causes DUA only in the **early phase**.

Bladder Outlet Obstruction

Rodent partial-BOO model — three stages:

obstruction (ring or ligature) → bladder distension → **compensation:** hypertrophy and hyperplasia, sharp weight gain then plateau, increased blood supply, near-normal function → variable interval → **decompensation:** falling contractility, poor emptying, reduced response to stimulation and agonists, **fibrosis**.

- **If obstruction is not relieved before decompensation and fibrosis, dysfunction is permanent.**
- **Obstructed bladders show reduced blood flow** (laser Doppler) and denervation in both humans and animals.

Ischaemia-reperfusion hypothesis:

raised intravesical pressure to overcome resistance → raised wall tension (**law of Laplace**) → vessel compression, ischaemia, hypoxia → reperfusion each fill-void cycle → reactive oxygen species and free

intracellular calcium → proteases, phospholipases, lipid peroxidation → damage to nerves, synaptic membranes, mitochondria, sarcoplasmic reticulum → **denervation and detrusor decompensation**.

- **Poor translation:** immature, often female animals with **acute** obstruction; the sequence is not seen with urethral stricture and does not explain female DUA.
- **Long-term follow-up of men with BOO (over a decade):** no significant fall in Pdet@Qmax, Qmax fell by only about 1 mL/s, and few needed intervention — prolonged BOO need not decompensate.
- **Most men with acute urinary retention have preserved detrusor function.** Some with chronic retention also do — typically with **poor compliance and upper-tract change (high-pressure chronic retention)**. Why outcomes diverge is unknown; the picture is probably multifactorial.

Diabetes Mellitus

- **Diabetes-induced bladder dysfunction (DBD)**, classically **diabetic cystopathy**: emptying efficiency falls in a time-dependent way over the disease course, by **combined neurogenic and myogenic** mechanisms.
- **Neurogenic:** autonomic neuropathy — axonal degeneration and segmental demyelination → **diminished sensation**. Driven by hyperglycaemia via the **polyol pathway, free radicals, protein kinase C activation and advanced glycation end products**.
- **Reduced nerve growth factor (NGF)** (maintains sympathetic and sensory nerves): low dorsal-root-ganglion NGF accompanies raised PVR and capacity in diabetic rats.
- **Myogenic:** altered intercellular connections, excitability, intracellular signalling and receptor density; animal data show both increased and decreased contractility.

DBD TEMPORAL THEORY (DANESHGARI) — EARLY VERSUS LATE

	Early (compensated)	Late (decompensated)
Driver	Hyperglycaemic osmotic diuresis → stretch, raised pressure, compensatory hypertrophy	Accumulated oxidative-stress products
Clinical picture	Storage symptoms	Poor sensation, impaired voiding, LUTS

- The theory reconciles the contradictory animal findings: contractility rises early and falls late.

Neurologic Disease or Injury

NEUROLOGICAL CAUSES OF DUA

Condition	Frequency / pattern of DUA	Key point
Stroke	About half retain acutely; up to 75% of these acontractile	"Cerebral shock" ; chronic sequela is neurogenic DO , not DUA
Parkinson disease (PD)	<20%; DO far commoner	Anticholinergic anti-Parkinson drugs contribute — none showed DUA once these were withdrawn before UDS
Multisystem atrophy (formerly Shy-Drager)	52–95% ; most develop retention	Atrophy of efferent parasympathetic nerves; often confused with PD
Multiple sclerosis	~20%	Especially with lumbosacral cord plaques
Lumbosacral cord, cauda equina, sacral/pelvic nerve injury	Common	Trauma or disc prolapse → cauda equina syndrome
Radical pelvic surgery	LUTD after radical hysterectomy ~72%; after rectal surgery now <5%	Injury to pelvic plexus (anterolateral lower rectal wall) or postganglionic fibres (lateral upper vaginal wall); fall with nerve-sparing; many recover by 1 year
Infective: Guillain-Barré, herpes zoster	—	Reversible

Condition	Frequency / pattern of DUA	Key point
Progressive: AIDS, neurosyphilis (tabes dorsalis)	—	Permanent ; tabes common pre-antibiotic era

- **True post-surgical DUA incidence is unclear** — few studies compare pre- and postoperative UDS. Recovery is attributed (feline pelvic plexus extirpation) to **restitution of intrinsic cholinergic nerves and myocyte regeneration** after initial degeneration.
- **Trap**: in stroke and PD the expected finding is **DO**; DUA is acute (stroke) or drug-related (PD). MSA is the degenerative condition that gives DUA and retention.

Pelvic Arterial Ischemia

- **Vascular risk factors** (ageing, smoking, diet, atherosclerosis, metabolic syndrome, hypertension, hyperlipidaemia, diabetes) may cause **chronic bladder ischaemia**, analogous to their role in erectile dysfunction.
- **Animal pelvic arterial insufficiency** → wall ischaemia, low oxygenation → **oxidative stress** → **ultrastructural damage and denervation** → features of DUA in some studies.

KEY POINTS: ETIOPATHOGENESIS

- Aetiopathogenesis is incompletely understood; diverse factors and mechanisms are implicated.
- **Confirmed**: neurological injury/disease and diabetes mellitus. **Probable, not definite**: BOO, ageing, atherosclerosis.
- These factors impair processes essential to an efficient voiding contraction.
- Mechanisms are **myogenic** (myocytes or ECM) or **neurogenic** (central control via afferent or efferent nerves).

Concept — *Every cause acts on one link of the voiding reflex loop — afferent, central, efferent or muscle — and afferent failure is the unifying theme of age-related DUA: the contraction stops early because the signal sustaining it fades.*

DIAGNOSIS

- **Invasive UDS is the only accepted way** to estimate detrusor voiding function clinically.
- **No universally agreed criteria** exist; current methods measure almost only **contraction strength**, neglecting **speed and sustainability**.

Detrusor Contraction Strength

- **A contraction generates pressure and flow**, so DUA is usually diagnosed by **low Qmax plus low Pdet@Qmax**. Male ranges come from men having outlet surgery; healthy ranges are poorly characterised.
- **Limitation 1 — underestimation**: the **bladder outlet relation (BOR)**, an adaptation of the **Hill equation**, makes pressure and flow inversely related during voiding (high flow, low pressure and vice versa), so **Pdet@Qmax is the point of lowest pressure**.
- **Limitation 2 — outlet resistance ignored**: with a low Pdet, low Qmax may reflect BOO; a normal Qmax may reflect a **low-resistance outlet** (e.g. post-prostatectomy incontinence).
- **Hence isovolumetric estimates** — by calculation or by interrupting flow; many are complex or impractical.

MEASURES OF DETRUSOR CONTRACTILITY COMPARED

Measure	Formula / method	Thresholds	Strengths	Weaknesses
Watts factor (WF)	$WF = [(P_{det} + a)(V_{det} + b) - ab] / 2\pi$; $a = 25 \text{ cm H}_2\text{O}$, $b = 6 \text{ mm/s}$; V_{det} = detrusor shortening velocity. Use WF_{max}	<7 W/m² reduced (expert opinion); some use 10 W/m ²	Power per unit bladder area; minimally volume-dependent, unaffected by BOO	Complex; no validated cut-offs ; little clinical use
Projected isovolumetric pressure (PIP), Schäfer	BOR drawn as straight line from $P_{det}@Q_{max}$ back to y-axis on Schäfer nomogram: $PIP = P_{det}@Q_{max} + K \cdot Q_{max}$. K = 5 in men with BPH; 1 in older women	>150 strong; 100–150 normal; 50–100 weak; <50 very weak	Quick; estimates isovolumetric pressure	K population-specific
Bladder contractility index (BCI), Abrams	$P_{det}@Q_{max} + 5Q_{max}$ (same as PIP, men)	>150 strong; 100–150 normal; <100 weak	Simple, easy to obtain	Male-derived; may not apply to other groups
Detrusor contraction coefficient (DECO)	$(P_{det}@Q_{max} + 5Q_{max}) / 100$	<1 weak	Simple	As BCI
Detrusor contractility parameter (DCP)	Hill plot of isovolumetric phase → maximum velocity of contractile-element shortening (VCE); uses t_{20–80} (time for pressure to rise 20–80% of maximum)	—	Uses the isovolumic phase	Max pressure depends on opening pressure → invalid with BOO ; experimental

- **PIP, BCI and DECO are one calculation**; they **overestimate** isovolumetric pressure (population adjustment needed) and are less reliable on retest than direct occlusion.
- **Figure 111.6:** $P_{det}@Q_{max}$ plotted on the Schäfer nomogram and projected to the y-axis; the example projects to 130 cm H₂O — **normal**.

DIRECT ISOVOLUMETRIC MEASUREMENT — STOP TESTS VERSUS CONTINUOUS OCCLUSION

	Stop test (voluntary / mechanical)	Continuous occlusion
Method	Flow interrupted after it begins — by the patient contracting the sphincter (voluntary) or by the investigator, e.g. tugging a catheter balloon against the bladder neck (mechanical)	Outflow blocked before and throughout the contraction
Accuracy	Voluntary and mechanical correlate well in both sexes; voluntary reads ~20 cm H₂O lower (reflex detrusor inhibition from sphincter contraction)	Greater test-retest reliability ; correlates with ability to empty
Sustainability	Not assessed	Assessed
Drawback	Voluntary test difficult or impossible with sphincter weakness	No representative flow for that contraction; painful ; research only

TABLE 111.2 SUMMARY OF DIAGNOSTIC METHODS AND CRITERIA — *from Osman, Chapple, Abrams et al., Eur Urol 2014*

Type	Methods	Advantages	Limitations
Mathematical calculation	Watts factor; detrusor shortening velocity	See comparison above; velocity may detect early DUA	Complex; unvalidated; no sustainability

Type	Methods	Advantages	Limitations
Indices	DECO; BCI	Simple	No sustainability; limited to derivation group
Occlusion testing	Voluntary / mechanical stop test; continuous occlusion	Real-time strength; no calculation	Painful; impractical; stop test underestimates; unusable in some groups
Ranges of urodynamic measurements	e.g. Pdet@Qmax <40 cm H ₂ O; Qmax <15 mL/s	Simple	No accepted normals; underestimate strength ; ignore coexisting BOO + DUA

- **What the table tests:** the trade-off between **practicality and validity** — the simple methods are the least accurate; the accurate ones are painful or impractical.
- **Criteria disagree:** BCI and PIP agree with each other, but applied to non-neurogenic LUTS the various criteria diagnose DUA in anywhere from **5% to 55%** — standardisation is needed.

Detrusor Contraction Speed

- **Slow contraction could cause symptoms**, but speed is **not in the ICS definition**.
- **Detrusor shortening velocity:** $V_{det} = (Q/2)[3/(V + V_t)/4\pi]^{0.66}$ (Q flow, V bladder volume, V_t non-contracting wall volume).
- **A fall in Vdet precedes a fall in WF** in longitudinal studies of both sexes — a potential early marker.

Detrusor Contraction Duration

- **In the ICS definition, but normal limits are undefined** and few studies measure it.
- **Paradox:** in men with BPH, unobstructed poor-contraction patients had **longer** contractions than unobstructed normals.
- **Duration probably reflects mechanism** — early reflex termination would shorten it. A standard measurement method is needed.

Bladder Sensation

- **Afferents initiate and sustain contraction;** men with DUA have **reduced sensation** versus other LUTDs.
- **Usual method: filling sensations** (first sensation, first desire, strong desire, capacity; normal thresholds exist) — **subjective:** reported even without filling, and vary with setting and anxiety.
- **Current perception threshold testing:** more objective, but invasive and unvalidated.

Ambulatory Urodynamics

- **Failure to void on UDS:** bashful bladder (poor pelvic floor relaxation + reflex detrusor inhibition) or true DUA / acontractility — **history usually separates them**.
- **If in doubt → ambulatory UDS:** over four-fifths of apparently acontractile patients contract. Criteria are **not standardised** and differ from conventional UDS.

KEY POINTS: DIAGNOSIS

- There is **no published consensus** on diagnostic criteria for DUA.
- Most criteria focus on contraction **strength** as determined by Pdet during UDS.
- Several indices estimate isometric strength; most are unvalidated and not applicable to all groups.
- **Stop and occlusion tests are more reliable** but may be painful and impractical.

- The significance of contraction speed and duration is unclear.

Concept — *Because pressure and flow trade off along the **bladder outlet relation**, no single voiding point measures strength; every index is an attempt to recover **isovolumetric pressure**, and each gains either practicality or validity at the expense of the other.*

MANAGEMENT

- **Goals:** symptoms and quality of life (QoL); prevent **urinary tract infection (UTI)**, **stones**, **reflux with upper-tract back-pressure**, **skin damage from overflow**.
- **Nothing effectively improves detrusor function** → **drainage** or **lower outlet resistance** (e.g. relax the external sphincter).

MANAGEMENT OPTIONS IN DUA (FIG. 111.7)

Aim	In routine clinical use	Experimental	Not recommended
Drainage	Timed voiding / double voiding; catheterisation; urinary diversion	—	Compression (Credé); reflex voiding; Valsalva voiding
↓ Outlet resistance	Pelvic floor physiotherapy and biofeedback; α -blockers; muscle relaxants; bladder outlet surgery	Intrasphincteric botulinum toxin	—
↑ Detrusor function	Sacral neuromodulation; anterior sacral root stimulator	Intravesical prostanoids; intravesical electrotherapy; detrusor myoplasty; reduction cystoplasty	Anticholinesterases; muscarinic agonists

- **Trap:** the drugs that sound most logical — **muscarinic agonists and anticholinesterases** — are in the "not recommended" column; the only routine detrusor-directed options are **neuromodulation and root stimulation**.

Initial Assessment

- **Routine urological evaluation:** bladder diary, digital rectal examination, urinalysis, uroflowmetry, ultrasound PVR.
- **Neurological assessment:** sacral dermatomes, anal tone, **bulbocavernosus reflex**, lower-limb reflexes; deficits → specialist referral, commonly **spinal MRI** (lumbar cord, cauda equina).
- **Drug history:** agents reducing contractility (anticholinergic, opioid; Box 111.1) or raising outlet resistance (**α -adrenoceptor agonists**).
- **Faecal impaction / constipation** can obstruct directly; treating it may improve emptying.

Conservative Management

- **Scheduled voiding** for sensory loss; **double voiding** to empty better and cut frequency.
- **Bladder expression (Valsalva, Credé):** only in specific neurogenic cases — **DUA with an incompetent sphincter**. Otherwise avoided: high vesical pressure → **vesicoureteric reflux or reflux into prostate and seminal vesicles**.
- **Pelvic floor physiotherapy and biofeedback:** effective in **dysfunctional voiding**, where a non-relaxing pelvic floor and sphincter obstruct flow and **reflexly inhibit the detrusor** — including some "acontractile" patients who contract on ambulatory UDS.

- **ISC is the preferred drainage for problematic high PVR**, given adequate cognition and dexterity: safe, effective, **fewer infections than indwelling catheters**.
 - **Problems:** urethral bleeding (about one-third), false passages, time and social restriction, psychological barriers (fear of harm or infection).
- **Avoid a long-term indwelling urethral catheter**; if ISC is impossible or declined, **suprapubic catheter is the best long-term option**.

Pharmacotherapies

DRUGS FOR DUA

Agent / class	Mechanism / target	Route	Evidence	Status
Bethanechol, carbachol (direct muscarinic agonists)	Quaternary amines; muscarinic-selective, not subtype-selective ; act on M_3	Oral	Meta-analysis (mainly postoperative retention): benefit in retention, no overall PVR difference ; evidence small and poor	Generally avoided
Distigmine (anticholinesterase)	Blocks acetylcholine breakdown; needs some endogenous acetylcholine	Oral	As above	Avoided
α-Blockers (tamsulosin, urapidil)	↓ Outlet resistance	Oral	Tamsulosin 0.2 mg improved voiding LUTS and Qmax in some women with DUA; urapidil alone: no Qmax gain, PVR fell in women only	Routine use
α-Blocker + parasympathomimetic (urapidil + bethanechol or distigmine)	↓ Resistance + ↑ contraction	Oral	Single-blind randomised trial: better flow in both sexes , lower PVR in women	Therapeutic option
Prostanoids (PGE_2 , $PGF_{2\alpha}$)	Enhance detrusor contraction	Intravesical	Mixed; pooled analysis of three randomised trials: more successful voiding (postoperative retention)	Experimental
ASP8302	Positive allosteric modulator of M_3 ; acts only when acetylcholine is present (voiding), inactive in storage	Oral	Randomised placebo-controlled: no difference in PVR (primary end point)	Negative
TAC-302	Cyclohexanoic long-chain fatty alcohol derivative; promotes neurite outgrowth	Oral	Phase 2 randomised: ↑ BCI in males and voiding efficiency in both sexes	Investigational

Parasympathomimetics for Underactive Bladder

- **Acetylcholine acting on M_3 receptors** is the principal mediator of bladder contraction.

WHO MIGHT RESPOND TO A PARASYMPATHOMIMETIC

Likely to help	Unlikely to help
Reduced or absent stimulus: reduced efferent input, impaired acetylcholine release, increased acetylcholine breakdown	Reduced tissue responsiveness: myocyte dysfunction, wall fibrosis; loss of detrusor muscle — no drug will work

- **Bethanechol raises wall stiffness and sensation;** responders show lower sensory thresholds → **intact sensory pathways may be a prerequisite**, explaining failure in unselected groups. Or doses are too low, limited by systemic effects.
- **Side effects:** nausea, **bronchospasm**, abdominal cramps, diarrhoea, salivation, flushing, visual disturbance; **rare but lethal cardiac depression and arrest.**

α-Adrenoreceptor Antagonists

- **Lower outlet resistance** in neurogenic bladder and dysfunctional voiding; results and combination therapy in the drug table.

Prostanoids

- **Five endogenous prostanoids** — PGE₂, PGF_{2α}, PGI₂, PGD₂, thromboxane A — are made in the bladder wall and released on stretch.
- **They enhance contraction to acetylcholine and adenosine triphosphate (ATP);** non-steroidal anti-inflammatory drugs abolish isolated-bladder tone, reversed by prostanoids. Efficacy in DUA is unclear.

PROSTANOID ACTIONS ON DETRUSOR AND URETHRA

Prostanoid	Detrusor	Urethra
PGE ₁ , PGE ₂	Contraction ↑	Relaxation
PGF _{2α}	Contraction ↑	Contraction

- **Side effect:** PGE and PGF also **stimulate uterine contraction.**

Novel Agents

- **ASP8302 negative, TAC-302 positive** in small randomised studies (drug table).

Future Prospects in Pharmacotherapy

- **Bladder-specific parasympathomimetic** allowing dose escalation — unlikely on current knowledge; **presynaptic M₂ antagonists** — promising.
- **Uroselective exogenous prostanoids** (sparing uterus and gut) in development; oral efficacy uncertain.
- **Transient receptor potential (TRP) agonists**, e.g. TRPV4 agonist GSK1016790A, increase contractility.
- **Other-organ analogues** (cardiac inotropes, gut prokinetics); **sensory sensitisers** for afferent failure.
- **Prerequisite:** better understanding of normal contraction and of DUA mechanisms.

Electrical Stimulation

ELECTRICAL STIMULATION TECHNIQUES IN DUA

Technique	Target / mechanism	Requirements / indication	Evidence and limits
Sacral anterior root stimulator (Brindley)	Stimulates anterior sacral roots via implanted receiver, wires, external transmitter	Complete spinal cord injury (SCI); intact peripheral efferents, no myogenic failure — shown by reflex contractions on filling preoperatively	Also fires urethral somatic efferents → intermittent stimulation → interrupted voiding ; Sauerwein added total sacral rhizotomy to abolish reflexes

Technique	Target / mechanism	Requirements / indication	Evidence and limits
Intravesical electrotherapy (IVE) (Katona)	Cathode in saline-filled bladder, skin electrode in sensate area; activates mechanosensitive Aδ afferents → central reflex detrusor activation; repeated use up-regulates the pathway	Some afferent and muscle function; fails with complete denervation or fibrosis ; mainly paediatric	Daily ≥1 h sessions, 10–15 as trial ; ~47 for durable effect in children; encouraging uncontrolled results, but the only sham-controlled randomised trial was negative and co-delivered biofeedback confounds; small UTI risk; resource-heavy
Sacral neuromodulation (SNM) (Tanagho and Schmidt)	Inhibits abnormal urethral afferents that block bladder afferents at the sacral cord → restores afferent flow to PAG and brain → sensation and contraction return; direct efferent effect less likely	Non-obstructive retention / DUA with sphincter or pelvic floor disorder; potential in DHIC	Good efficacy; needs some detrusor function; key unresolved question is coexistent DUA + BOO
Transcutaneous electrotherapy	Skin electrodes	Children	Small randomised trial versus urotherapy: improved voiding time, capacity, flow; no sham arm

- **Anterior root stimulation exploits the slower relaxation of smooth than skeletal muscle:** bursts keep the detrusor contracted while the sphincter relaxes between bursts.
- **DHIC is hard to treat** because treating the DO worsens the DUA and vice versa — hence SNM's appeal.

Concept — *Each stimulation method needs a surviving link in the loop: Brindley needs efferents and muscle, IVE and SNM need afferents and muscle; none can restore a fibrotic or denervated bladder.*

Botulinum Toxin

- **Botulinum neurotoxin A (BoNT-A) into the urethral sphincter** lowers outlet resistance in **detrusor-sphincter dyssynergia**.
- **Rationale in DUA:** relax the sphincter to **remove reflex detrusor inhibition**, or to facilitate Valsalva voiding.
- **Technique:** **50 U in 4 mL** saline (or 100 U in 8 mL) — cystoscopic in males, periurethral in females.
- **Results:** some catheter-dependent patients voided with low PVR, ISC fell; about half of idiopathic cases improved.
- **Predictors of response:** **better bladder sensation; poor sphincter relaxation** responds best, pure DUA or DHIC about one-third.
- **Limits:** short-lived; no dose studies; **unlicensed — clinical trials only**.

Surgery

Bladder Outlet Surgery

- **Controversial in males with non-neurogenic DUA;** no high-level evidence.
- **Meta-analysis of comparative studies of transurethral surgery:** DUA → **smaller improvement in International Prostate Symptom Score (I-PSS) and Qmax**; QoL change and PVR similar to preserved contractility.

BLADDER OUTLET SURGERY — THE TWO CLINICAL SCENARIOS

	DUA, low PVR, symptom-driven	DUA, catheter-dependent (chronic retention)
Reasoning	Emptying with a low Pdet implies BOO is unlikely to be significant	Later decompensation; aim is to lower resistance enough to empty
Evidence	TURP followed long term (~11 years): no symptomatic improvement ; small, doubtful fall in BOOI (= Pdet@Qmax – 2Qmax) with no gain in flow or voiding efficiency — no better than conservative care	Even less evidence; flow and PVR rarely normalise, but a significant proportion resume voiding
Verdict	Surgery unlikely to help	Consider in younger, fit patients wanting to be catheter-free

- **Predictors of poor outcome in retention:** voiding pressure <45 cm H₂O, age >80, PVR >1500 mL.
- **Females: urethral dilatation** lacks evidence; **bladder neck resection or incision not recommended** — incontinence or bladder neck stenosis.

Concept — *Outlet surgery helps only where outlet resistance is the part of the problem that can be changed: the symptomatic low-PVR man has little to gain, while the catheter-dependent man may gain enough to void.*

Urinary Diversion

- **Rarely needed outside neurogenic patients**, since ISC and indwelling catheters are acceptable and safe.
- **Continent catheterisable stoma** when urethral ISC is impossible and a suprapubic catheter is declined — typically **young women with idiopathic retention refractory to SNM**.
- **Incontinent diversion (ileal conduit)** occasionally, for **renal deterioration** from intramural ureteric obstruction by a thick-walled bladder.

Reconstructive Surgery

RECONSTRUCTIVE AND REGENERATIVE OPTIONS

Procedure	Technique	Outcome	Status
Latissimus dorsi detrusor myoplasty (LDDM) (Stenzl)	Free muscle; pedicle to inferior epigastric vessels , nerve to intercostal branch ; spiral wrap over >three-quarters of the bladder, anchored to pelvic floor fascia and ligaments	Catheter-dependent acontractile patients: most voided with low PVR; BCI rose to the strong range ; complications (thromboembolism, abscess, wound infection) in one-third, none long term	Investigational
Rectus abdominis detrusor myoplasty	Less morbid, simpler than LDDM	Small SCI series: complete spontaneous voiding, no major complications at over 1 year	Investigational
Partial cystectomy / reduction cystoplasty	Excise excess bladder	Few old series	Abandoned
Tissue-engineered bladder	Engineered whole bladder	Storage only — innervation for volitional emptying cannot be recreated	Research
Autologous muscle-derived stem cells (AMDSC)	Injected into the bladder wall	Two open-label single-arm studies: improvement in the majority, higher voiding efficiency; in a small male series urodynamic gains and most stopped ISC	Research

Regenerative Medicine

- **Animal stem cell studies** (DUA model → injection into bladder or neural injury site → urodynamics): several show improved contractility.

- **Barriers:** incomplete understanding of lower urinary tract structure and function, and whether injected cells can **acquire functional innervation** and contribute meaningfully.

KEY POINTS: MANAGEMENT

- There are **no effective pharmacotherapies** for DUA.
- Parasympathomimetics have questionable efficacy, potentially **fatal** systemic effects, and are not used routinely.
- Intravesical electrotherapy shows promise in restoring contraction but is time- and resource-consuming.
- **SNM is effective** in restoring voiding in DUA associated with urethral sphincter / pelvic floor disorders.
- Outlet surgery has a **high failure risk in men not reliant on catheters**. Catheter-dependent men with DUA fare worse than those with normal detrusors but **may still benefit**.

Concept — *Treatment works only where some part of the loop survives* — **intact sensation is the recurring predictor of response** (*bethanechol, botulinum toxin, IVE*); *where muscle is lost, only drainage or replacement of the muscle remains*.

CONCLUSIONS

- **DUA is not uncommon and hard to treat**, and has been under-researched compared with OAB and DO.
- **A symptom-based definition is difficult** — wide symptom spectrum, diverse conditions, no accurate non-invasive marker.
- **Highly prevalent in both sexes presenting with LUTS**; causes affect every part of the micturition reflex.
- **Preserved bladder sensation predicts better response to treatment**.
- **Drugs to enhance strength, duration or sensation need mechanistic understanding first**; regenerative, stem cell and gene therapy depend on the same.